

Pair-Bound Lyapunov Stability for a Scalar Dissipative Kuramoto-Type Mean-Field Equation, Formalized in Lean 4

Abstract

We establish global asymptotic stability of the partially locked state for a scalar dissipative mean-field equation that shares the self-consistency structure of the Kuramoto–Ott–Antonsen system. The central device is a *pair bound*: a pointwise algebraic inequality, lifted to the continuum via Fubini, that renders the L^2 Lyapunov function non-increasing without recourse to spectral theory or Fourier analysis. Combining this with a dominated-convergence bootstrap that prevents the order parameter from vanishing, we obtain convergence $r(t) \rightarrow r^*$ from *all* initial data—no basin-of-attraction assumption is needed.

The entire argument is formalized in Lean 4 (Mathlib v4.30.0-rc1) with no `sorry` and no project-specific axioms. A conditional extension to the standard complex Ott–Antonsen equation and the full Kuramoto PDE is also provided, relying on two cited external results (nonlinear Landau damping and OA manifold attractivity). To the best of our knowledge, this constitutes the first machine-checked proof of a stability theorem for a Kuramoto-type model.

Main contributions.

- (1) A novel *pair bound* (Proposition 2.2) giving $\dot{V} \leq 0$ for the scalar dissipative model (2), with no antecedent in the Kuramoto literature.
- (2) Global stability without a basin condition: a DCT bootstrap (Proposition 2.10) shows $r(t) \geq r_{\min} > 0$ for all $K > K_c$, eliminating the assumption $V(0) < (r^*)^2$.
- (3) A complete machine-checked formalization covering Gaussian, Student- t ($\nu > 1$), compactly supported, and mixture distributions satisfying $\int \gamma d\mu < \infty$.
- (4) A conditional bridge to the standard complex OA and full Kuramoto PDE via rotation cancellation and two external axioms.

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1 Setup and statement of results

1.1 The standard complex Ott–Antonsen equation

On the Ott–Antonsen manifold, the Kuramoto–Sakaguchi PDE reduces to

$$\dot{z}(\omega, t) = -i\omega z + \frac{K}{2}(\bar{\eta} - \eta z^2), \quad \eta(t) = \int_{\mathbb{R}} \overline{z(\omega, t)} g(\omega) d\omega, \quad (1)$$

where $z(\omega, t) \in \mathbb{D}$ and $\eta(t) \in \mathbb{C}$ is the complex order parameter [5]. When g is symmetric and the initial data satisfy $z(\omega, 0) = \overline{z(-\omega, 0)}$, this symmetry persists and $\eta(t) = r(t) \in \mathbb{R}$. For the Lorentzian $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, residue calculus collapses (1) to the scalar ODE $\dot{r} = (K/2 - \gamma)r - (K/2)r^3$.

1.2 A scalar dissipative model

We study the scalar real mean-field equation

$$\dot{\alpha}(\omega, t) = -\gamma(\omega) \alpha + \frac{K}{2} r(t)(1 - \alpha^2), \quad r(t) = \int_{\mathbb{R}} \alpha(\omega, t) g(\omega) d\omega, \quad (2)$$

with $\alpha(\omega, t) \in (0, 1)$ and dissipation rate $\gamma(\omega) > 0$. We write $d\mu = g d\omega$ for the probability measure induced by the density g .

This is a *distinct* dynamical system from (1): the dissipative term $-\gamma\alpha$ replaces the rotational term $-i\omega z$. What the two systems share is the self-consistency equation defining the partially locked state.

Remark 1.1 (Scope). *Equation (2) is **not** obtained from (1) by restricting to real z ; the subspace $\text{Im}(z) = 0$ is not invariant under (1). The Lyapunov argument in Sections 2–2.5 applies to (2) directly. The conditional extension to (1) appears in Section 6.*

Model	Eq.	Uncond.?	Lean status	Relation to Kuramoto
Scalar dissipative	(2)	Yes	complete	Shares PLS self-consistency
Lorentzian ODE	Bernoulli	Yes	complete	Classical OA for Lorentzian g
n -pole	Finite	Yes	complete	Discrete analogue
Complex OA	(1)	No (1 axiom)	complete	Standard OA reduction
Full Kuramoto PDE	Kinetic	No (2 axioms)	complete	Original model

1.3 The partially locked state (PLS)

We assume the existence of a unique positive equilibrium (α^*, r^*) with $r^* \in (0, 1)$ satisfying

$$r^* = \int_{\mathbb{R}} \alpha^*(\omega) g(\omega) d\omega \quad (3)$$

and the pointwise balance

$$\gamma(\omega) \alpha^*(\omega) = \frac{K}{2} r^* (1 - \alpha^*(\omega)^2) \quad \text{for } g\text{-a.e. } \omega. \quad (4)$$

For the Lorentzian this holds whenever $K > K_c := 2\gamma$.

1.4 Main theorems

Theorem 1.2 (Basin stability). *Let $\gamma > 0$ with $\int \gamma d\mu < \infty$. Let (α^*, r^*) satisfy (4)–(3), and let (α, r) solve (2) with:*

- (i) $r(t) = \int \alpha(\omega, t) d\mu$ and $\alpha(\omega, t) \in (0, 1)$ for all $\omega, t \geq 0$;
- (ii) $V(0) := \int (\alpha(\omega, 0) - \alpha^*(\omega))^2 d\mu < (r^*)^2$;
- (iii) for each $M > 0$, there exists $\delta_0 > 0$ with $\alpha(\omega, 0) \geq \delta_0$ on $\{\gamma \leq M\}$.

Then $r(t) \rightarrow r^*$ as $t \rightarrow +\infty$.

Remark 1.3. *No persistence hypothesis appears among the assumptions. The proof derives order-parameter persistence ($r(t) \geq r_{\min} > 0$) from V antitonicity and Cauchy–Schwarz, then body persistence from the ODE barrier. Conditions (ii)–(iii) hold for any initial datum in a neighborhood of the PLS.*

Theorem 1.4 (Global stability). *Assume additionally that $\int (1/\gamma) d\mu < \infty$ and $K > K_c := 2/\int (1/\gamma) d\mu$. Then the conclusion of Theorem 1.2 holds **without** condition (ii): $r(t) \rightarrow r^*$ for all initial data satisfying (i) and (iii).*

Remark 1.5. *The threshold $K > K_c$ does not alone guarantee existence of (α^*, r^*) ; that is an additional hypothesis, verified for specific families (Gaussian, Student- t) via the implicit function theorem on (3).*

2 Proof of basin stability

The proof consists of six steps, presented in the order that the formal verification assembles them.

2.1 Step 1: Leibniz rule

Define the L^2 Lyapunov function

$$V(t) := \int_{\mathbb{R}} (\alpha(\omega, t) - \alpha^*(\omega))^2 d\mu. \quad (5)$$

Proposition 2.1 (Leibniz). *Under the hypotheses of Theorem 1.2, V is continuous and*

$$V'(t) = \int_{\mathbb{R}} 2(\alpha - \alpha^*) \dot{\alpha} d\mu \quad \text{for all } t > 0.$$

Proof. The pointwise derivative $\partial_t(\alpha - \alpha^*)^2 = 2(\alpha - \alpha^*)\dot{\alpha}$ satisfies $|2(\alpha - \alpha^*)\dot{\alpha}| \leq 2(\gamma(\omega) + K)$, since $|\alpha - \alpha^*| \leq 1$ and $|\dot{\alpha}| \leq \gamma + K$ on $(0, 1)$. The right-hand side is μ -integrable by hypothesis, so differentiation under the integral sign applies. $\square$

2.2 Step 2: The pair bound

Proposition 2.2 (Pair bound). *For any $\alpha_j, \alpha_k, \alpha_j^*, \alpha_k^* \in (0, 1)$:*

$$\alpha_j^*(\alpha_k - \alpha_k^*)^2 \left(\alpha_k + \frac{1}{\alpha_k^*} \right) + \alpha_k^*(\alpha_j - \alpha_j^*)^2 \left(\alpha_j + \frac{1}{\alpha_j^*} \right) \geq (\alpha_j - \alpha_j^*)(\alpha_k - \alpha_k^*)(2 - \alpha_j^2 - \alpha_k^2). \quad (6)$$

Proof. Write $p_j = \alpha_j - \alpha_j^*$, $p_k = \alpha_k - \alpha_k^*$. Multiplying both sides by $\alpha_j^* \alpha_k^* > 0$ and expanding:

$$\alpha_j^* \alpha_k^* (\text{LHS} - \text{RHS}) = (\alpha_j^* p_k - \alpha_k^* p_j)^2 + (\alpha_j^*)^2 \alpha_k \alpha_k^* p_k^2 + (\alpha_k^*)^2 \alpha_j \alpha_j^* p_j^2 + \alpha_j^* \alpha_k^* (\alpha_j^2 + \alpha_k^2) p_j p_k. \quad (7)$$

When $p_j p_k \leq 0$: each LHS summand is non-negative (factors involving $\alpha + 1/\alpha^*$ are positive) while the RHS is non-positive (since $2 - \alpha_j^2 - \alpha_k^2 > 0$).

When $p_j p_k > 0$: the first three terms of (7) are manifestly non-negative. The fourth is positive since $\alpha_j^2 + \alpha_k^2 > 0$ and $p_j p_k > 0$, so $\alpha_j = \alpha_j^* + p_j > 0$ forces every factor positive. $\square$

Proposition 2.3 (Continuum pair bound). *The double integral*

$$\iint \left[\alpha_1^*(\alpha_2 - \alpha_2^*)^2 \left(\alpha_2 + \frac{1}{\alpha_2^*} \right) + \alpha_2^*(\alpha_1 - \alpha_1^*)^2 \left(\alpha_1 + \frac{1}{\alpha_1^*} \right) - (\alpha_1 - \alpha_1^*)(\alpha_2 - \alpha_2^*)(2 - \alpha_1^2 - \alpha_2^2) \right] d\mu_1 d\mu_2 \geq 0,$$

where we abbreviate $\alpha_i = \alpha(\omega_i, t)$ and $\alpha_i^* = \alpha^*(\omega_i)$.

Proof. The integrand is pointwise non-negative by Proposition 2.2; integrate. $\square$

Proposition 2.4 (Monotonicity). *$V'(t) \leq 0$ for all $t > 0$.*

Proof sketch. Substituting (2) into the Leibniz formula and using the equilibrium relation (4), the integrand splits into three terms: one proportional to $(r - r^*)^2$ (controlled by Cauchy-Schwarz), one involving the quadratic form $Q = \int (\alpha - \alpha^*)^2 (\alpha + 1/\alpha^*) d\mu$, and a cross-term involving $S = \int (\alpha - \alpha^*)(1 - \alpha^2) d\mu$. The continuum pair bound (Proposition 2.3) supplies the key inequality $DS \leq r^* Q$, from which $V' \leq 0$ follows algebraically. $\square$

2.3 Step 3: V is antitone

Proposition 2.5. *V is non-increasing on $[0, \infty)$.*

Proof. Immediate from $V' \leq 0$ and continuity. $\square$

2.4 Step 4: Body–tail decomposition and Barbalat

Proposition 2.6 (Convergence). $V(t) \rightarrow 0$ as $t \rightarrow +\infty$.

Proof. Fix $\varepsilon > 0$. Decompose $V = V_{\text{body},M} + V_{\text{tail},M}$ along the partition $\{\gamma \leq M\} \cup \{\gamma > M\}$.

Tail. Since $(\alpha - \alpha^*)^2 \leq 1$ and $\int \gamma \, d\mu < \infty$, Markov's inequality gives $V_{\text{tail},M} \leq \mu(\gamma > M) \leq (1/M) \int \gamma \, d\mu \rightarrow 0$. Choose M_0 so that $V_{\text{tail},M_0} < \varepsilon/2$ uniformly in t .

Body. On $\{\gamma \leq M_0\}$, body persistence (derived from the basin condition via Cauchy–Schwarz and the ODE barrier) yields $\alpha(\omega, t) \geq \delta > 0$. The resulting coercivity

$$-V'(t) \geq c V_{\text{body},M_0}(t), \quad c = K\delta\delta^* \mu(\gamma \leq M_0) > 0,$$

forces $\int_0^\infty V_{\text{body},M_0} \, dt < \infty$. Since the time-derivative of V_{body,M_0} is bounded (by $2(M_0 + K)$), Barbalat's lemma gives $V_{\text{body},M_0}(t) \rightarrow 0$.

Combining: $V(t) < \varepsilon$ for t large enough. $\square$

2.5 Step 5: Cauchy–Schwarz closing

Proposition 2.7. $V(t) \rightarrow 0$ implies $r(t) \rightarrow r^*$.

Proof. $(r - r^*)^2 = \left(\int (\alpha - \alpha^*) \, d\mu\right)^2 \leq \int (\alpha - \alpha^*)^2 \, d\mu = V(t) \rightarrow 0$. $\square$

Proof of Theorem 1.2. Compose the five steps: Leibniz (Prop. 2.1) $\rightarrow$ pair bound and monotonicity (Props. 2.2–2.4) $\rightarrow$ antitonicity (Prop. 2.5) $\rightarrow$ body–tail convergence (Prop. 2.6) $\rightarrow$ Cauchy–Schwarz (Prop. 2.7). Order-parameter persistence and body persistence are derived internally from the basin condition (ii) and never assumed. $\square$

2.6 Step 6: Pair rigidity

Although not required for convergence, the pair bound also characterizes the set $\{V' = 0\}$.

Proposition 2.8. *The pair integrand (6) vanishes if and only if $\alpha_j = \alpha_j^*$ and $\alpha_k = \alpha_k^*$.*

Proof. When $p_j p_k > 0$, terms T_2 and T_3 in (7) are strictly positive unless $p_j = p_k = 0$. When $p_j p_k \leq 0$, each summand of the LHS of (6) is non-negative and must vanish individually, again forcing $p_j = p_k = 0$. $\square$

Remark 2.9 (LaSalle interpretation). *Proposition 2.8 shows $\{V' = 0\} = \{\alpha = \alpha^*\}$, the LaSalle characterization. An alternative proof path uses scalar asymptotic autonomy of each oscillator to deduce pointwise convergence $\alpha(\omega, t) \rightarrow \alpha^*(\omega)$, then dominated convergence for $V \rightarrow 0$.*

2.7 Removing the basin condition

The following proposition is the key additional ingredient for Theorem 1.4.

Proposition 2.10 (Uniform positivity of the order parameter). *Under hypotheses (i) and (iii) of Theorem 1.2, if $K > K_c := 2/\int(1/\gamma) \, d\mu$, then there exists $r_{\min} > 0$ with $r(t) \geq r_{\min}$ for all $t \geq 0$.*

Proof. The argument proceeds by contradiction in three steps.

Step A (DCT extraction). Define $G_n(\omega) := \min((n+1)\alpha(\omega, 0), K/(2\gamma(\omega) + K/(n+1)))$. Each G_n is dominated by the integrable function $K/(2\gamma)$, and $G_n \rightarrow K/(2\gamma)$ pointwise. By the dominated convergence theorem, $\int G_n \, d\mu \rightarrow (K/2) \int (1/\gamma) \, d\mu > 1$. Extract N with $\int G_N \, d\mu > 1$ and set $\varepsilon_0 := 1/(N+1)$. A short computation shows $\int \min(\alpha(\omega, 0), K\varepsilon_0/(2\gamma + K\varepsilon_0)) \, d\mu > \varepsilon_0$.

Step B (Self-improvement). Suppose $r(s) \geq \varepsilon_0$ on $[0, T]$. The ODE barrier then gives $\alpha(\omega, T) \geq \min(\alpha(\omega, 0), K\varepsilon_0/(2\gamma(\omega) + K\varepsilon_0))$ for each ω . Integrating: $r(T) > \varepsilon_0$.

Step C (Contradiction). If r ever dips below ε_0 , let $T = \inf\{t : r(t) \leq \varepsilon_0\}$. By continuity $T > 0$ and $r(T) = \varepsilon_0$, while $r \geq \varepsilon_0$ on $[0, T]$. Step B gives $r(T) > \varepsilon_0$, a contradiction. $\square$

Proof of Theorem 1.4. Proposition 2.10 provides $r \geq r_{\min} > 0$. This replaces the Cauchy–Schwarz bound from the basin condition in the body-persistence argument: on $\{\gamma \leq M\}$, the ODE barrier with $r \geq r_{\min}$ yields $\alpha(\omega, t) \geq \delta(M) > 0$. The body–tail decomposition and Barbalat’s lemma then give $V \rightarrow 0$, and Proposition 2.7 closes. $\square$

3 The Lorentzian case

For the Lorentzian $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, the full continuum system collapses to

$$\dot{r} = \left(\frac{K}{2} - \gamma\right)r - \frac{K}{2}r^3, \quad (8)$$

with equilibrium $r^* = \sqrt{1 - 2\gamma/K}$ for $K > 2\gamma$.

Theorem 3.1 (Lorentzian convergence). *For $K > 2\gamma > 0$ and any $r_0 \in (0, 1)$, the solution of (8) satisfies $r(t) \rightarrow r^*$.*

Proof. The substitution $w = r^{-2}$ transforms (8) into a linear ODE with explicit solution. Alternatively: $\dot{r} = (K/2)r((r^*)^2 - r^2)$ shows r is monotone and bounded, hence convergent. The limit satisfies $0 = (K/2 - \gamma)L - (K/2)L^3$; since $r > 0$, we obtain $L = r^*$. $\square$

This proof is fully end-to-end in Lean: the explicit Bernoulli formula constructs the solution, verifies the ODE, and gives convergence with no external inputs beyond the physical parameters (K, γ, r_0) .

4 Lean 4 formalization

4.1 Overview

The formalization spans 255 Lean 4 files (Mathlib v4.30.0-rc1). The core theorems depend on no project-specific axioms and contain no `sorry`. The development uses only Lean 4’s standard foundations (propositional extensionality, quotient soundness, and the axiom of choice).

There are three tiers of completeness:

Tier 1 (Lorentzian). Fully end-to-end: takes only (K, γ, r_0) and constructs solution existence, all required properties, and convergence. No assumed fields.

Tier 2 (Finite first moment). Derives everything from the system definition: Leibniz rule, pair bound, antitonicity, persistence, body–tail convergence, and Cauchy–Schwarz closing. The only inputs are $V(0) < (r^*)^2$ (for basin stability) or $K > K_c$ (for global stability) and body coherence at $t = 0$.

Tier 3 (n -pole). Finite-dimensional systems with explicit exponential decay rate $\dot{V} \leq -\mu V$, $\mu = Kc_{\min}\delta\delta^*$. Global stability for any $r(0) > 0$.

4.2 Correspondence table

Proposition	Lean declaration	File
Leibniz (Prop. 2.1)	<code>leibniz_oa_lyapunov</code>	<code>GeneralGMainTheorem</code>
Pair bound (Prop. 2.2)	<code>pair_bound</code>	<code>L2Lyapunov</code>
Cont. pair (Prop. 2.3)	<code>continuum_pair_nonneg</code>	<code>ContinuumLyapunov</code>
Monotonicity (Prop. 2.4)	<code>hV_rate</code>	<code>ContinuumUniformRate</code>
Antitonicity (Prop. 2.5)	<code>lyapunov_antitone</code>	<code>GeneralGMainTheorem</code>
Convergence (Prop. 2.6)	<code>coercive_drop_from_persistence</code>	<code>ContinuumBarbalat</code>
$r \rightarrow r^*$ (Prop. 2.7)	<code>sq_integral_le_integral_sq</code>	<code>GeneralGMainTheorem</code>
Rigidity (Prop. 2.8)	<code>pair_eq_zero_iff</code>	<code>ContinuumLyapunov</code>
Uniform positivity (Prop. 2.10)	<code>r_stays_positive_supercritical</code>	<code>ContinuumInstability</code>

Key Mathlib ingredients:

- differentiation under the integral sign (dominated convergence variant);
- monotone functions on $[0, \infty)$ from derivative bounds;
- non-negativity of integrals from pointwise non-negativity.

4.3 Nine independent proof paths

The formalization contains nine sorry-free proof paths addressing distinct mathematical angles (gap exclusion, IVT trapping, scalar asymptotic autonomy, Gronwall comparison, instability escape, etc.). Each provides an independent verification of basin stability or a closely related statement, offering redundancy against logical errors in any single approach.

4.4 Additional formalized results

Beyond stability:

- **Trifurcation:** $K < K_c \Rightarrow r \rightarrow 0$; $K = K_c \Rightarrow r \rightarrow 0$ at $O(1/\sqrt{t})$; $K > K_c \Rightarrow r \rightarrow r^*$.
- **Bifurcation monotonicity:** $r^*(K)$ is increasing for $K > K_c$.
- **Instability of incoherence:** via Chetaev's theorem for $K > K_c$.

5 The n -pole case

For a discrete frequency distribution $\sum_{k=1}^n c_k \delta_{\omega_k}$, the Lyapunov function $V = \sum_k c_k (\alpha_k - \alpha_k^*)^2$ satisfies a stronger estimate.

Theorem 5.1 (Exponential decay). *For any n -pole system with $K > K_c$, there exist $c_{\min}, \delta, \delta^* > 0$ such that*

$$\dot{V} \leq -K c_{\min} \delta \delta^* \cdot V.$$

Proof outline. Differentiate, apply the finite pair bound, and use the weight floor $c_{\min} = \min_k c_k$ to extract a uniform coercivity constant. The exponential rate is sharp: it degenerates as $c_{\min} \rightarrow 0$ in the continuum limit, where Barbalat's lemma replaces the Gronwall bound. $\square$

Remark 5.2. *For n -pole systems, persistence is automatic in finite dimensions and the basin condition is not needed: any $r(0) > 0$ converges. For the continuum model, the DCT bootstrap (Proposition 2.10) closes the same gap.*

6 Complex OA: rotation cancellation and conditional stability

The pair-bound technique of Section 2 is specific to the scalar dissipative model (2). We now examine which structural elements carry over to the standard complex OA (1).

6.1 The Dietert energy

Proposition 6.1. *The functional $\Psi(t) := -\int \log(1 - |z|^2) d\mu$ satisfies $\dot{\Psi} = K |\eta|^2 \geq 0$.*

Proof. Differentiate $-\log(1 - |z|^2)$ using $d|z|^2/dt = K \operatorname{Re}(\eta z)(1 - |z|^2)$ from the complex OA. The integrand reduces to $K \operatorname{Re}(\eta z)$; integrating and using $\bar{\eta} = \int z g$ yields $\dot{\Psi} = K \operatorname{Re}(\eta \bar{\eta}) = K |\eta|^2$. $\square$

6.2 Rotation cancellation

The L^2 error $V(t) = \int |z - z^*|^2 d\mu$ has derivative

$$V' = 2 \int \operatorname{Re}(\overline{(z - z^*)} \cdot \dot{z}) d\mu.$$

After subtracting the equilibrium equation:

Proposition 6.2 (Rotation cancellation). $\operatorname{Re}(\overline{(z - z^*)} \cdot (-i\omega(z - z^*))) = 0$.

Proof. $-i\omega|z - z^*|^2$ is purely imaginary. $\square$

The rotation cancellation removes the $-i\omega$ term from V' , but the remaining nonlinear coupling—involving $\eta \in \mathbb{C}$ rather than $r \in \mathbb{R}$, and the quadratic terms $\eta^*(z^{*2} - z^2)$ —does not factor into the scalar pair-bound structure. Convergence is instead obtained from nonlinear Landau damping [9]: for perturbations of Sobolev regularity, phase mixing gives algebraic decay, stated as an axiom in the formalization.

6.3 Full Kuramoto PDE

Theorem 6.3 (Conditional bridge). *For the Kuramoto–Sakaguchi PDE with analytic symmetric g and $K > K_c$: if $\|\eta_{\text{OA}}\| \rightarrow r^*$ (via Landau damping) and $\|\eta_{\text{full}} - \eta_{\text{OA}}\| \rightarrow 0$ (via OA attractivity), then $\|\eta_{\text{full}}\| \rightarrow r^*$.*

Proof. Triangle inequality and squeeze. $\square$

Remark 6.4. *Both hypotheses are formalized as Lean axioms with opaque, non-vacuous conditions. Discharging them requires formalizing the spectral analysis of [9, 2], which remains future work.*

7 Discussion

7.1 Summary of results

Theorem	Model	Hypotheses	Conclusion	Sorry	Axioms
Basin (Thm. 1.2)	Scalar dissipative	$\int \gamma < \infty, V(0) < (r^*)^2$	$r \rightarrow r^*$	0	0
Global (Thm. 1.4)	Scalar dissipative	$\int \gamma + \int 1/\gamma < \infty, K > K_c$	$r \rightarrow r^*$	0	0
Lorentzian (Thm. 3.1)	Bernoulli ODE	$K > 2\gamma, r_0 \in (0, 1)$	$r \rightarrow r^*$	0	0
n -pole (Thm. 5.1)	Discrete	$K > K_c, r(0) > 0$	$r \rightarrow r^*$	0	0
Bridge (Thm. 6.3)	Complex OA + PDE	Analytic $g, K > K_c$	$\ \eta\ \rightarrow r^*$	0	2

7.2 Significance

The critical coupling K_c , the equilibrium profile α^* , and the order parameter r^* are *identical* between the scalar dissipative model and the standard Kuramoto–OA system restricted to the positive real subspace. Theorem 1.4 therefore establishes that this shared steady state is a global attractor for a scalar dissipative flow with the Kuramoto phase-transition threshold, the Kuramoto fixed point, and the Kuramoto self-consistency structure.

The pair bound is a property of the self-consistency *algebra*—the equilibrium relation (4) and the integral constraint (3)—rather than of the specific dynamics. This suggests potential transferability to other mean-field models with the same equilibrium structure.

7.3 What is new

1. **The pair bound.** A pointwise algebraic inequality lifted to a continuum double integral via Fubini, giving $\dot{V} \leq 0$ without spectral theory, Fourier analysis, or Volterra equations. Prior Kuramoto stability arguments relied on one of these [1, 2, 6].
2. **Machine-checked proof.** The first Lean formalization of a stability theorem for a Kuramoto-type model, to the best of our knowledge.
3. **Rotation cancellation.** The structural observation that $-i\omega$ drops from the complex OA Lyapunov derivative (Proposition 6.2), relevant to the conditional extension.
4. **Finite first moment coverage.** The condition $\int \gamma d\mu < \infty$ (with $\gamma \lesssim 1 + |\omega|$) covers Student- t with $\nu > 1$ —including the infinite-variance regime $1 < \nu \leq 2$ —Gaussian, compact support, and mixtures.

7.4 Limitations

1. **Scope.** The unconditional results apply to the scalar dissipative model (2), not to the standard complex OA (1). The two share their equilibrium structure but differ in their transient dynamics.
2. **Conditional axioms.** The complex OA and PDE results rely on two unprovable (within this formalization) axioms encoding Landau damping and OA attractivity. Their conditions are opaque Lean propositions that cannot be trivially discharged.
3. **Type-level precision.** The Lean conclusion states $\text{Re}(\eta)^2 \rightarrow (r^*)^2$. On the symmetric subspace $\eta \in \mathbb{R}$ (proved), so $\text{Re}(\eta)^2 = |\eta|^2$; but the formal statement carries this minor cosmetic asymmetry.

7.5 Open problems

1. Formalize nonlinear Landau damping (currently an axiom citing [9]).
2. Formalize OA manifold attractivity (currently an axiom citing [2]).
3. Extend to distributions without finite first moment (Student- t with $\nu \leq 1$) beyond the Lorentzian case.
4. Remove the basin condition for the standard complex OA without appealing to Landau damping.

Appendix: Reproducibility

Repository	<code>https://github.com/taejun-song/kuramoto-lean</code>
Commit	<code>e94219b</code>
Lean toolchain	<code>leanprover/lean4:v4.30.0-rc1</code>
Mathlib revision	<code>0c154d67</code> (Mathlib4)
Build command	<code>lake build</code>
Axiom check	<code>#print axioms real_scalar_complete</code> <code>outputs only propext, Quot.sound, Classical.choice</code>
Sorry check	<code>grep -rn "sorry" KuramotoLean/RealScalarComplete.lean</code> returns no matches

Unrelated `sorry` instances exist in `NeuralMeanField.lean` (2) and `MeanFieldLimit.lean` (4), both outside the dependency closure of any theorem above, verifiable via `#print axioms`.

References

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